📊 Executive Portfolio Strategy & Liquidation Dashboard PoC

Author: Gram99
Target Stakeholders: Risk Modeling, Asset Managers, Recovery Operations & Strategy (First & Second Lines of Defense)

A sophisticated **Proof of Concept (PoC)** tool designed for Financial Risk Modelers and Asset Managers to visualize, simulate, and strategize the lifecycle of a credit portfolio.

This interactive Streamlit dashboard transforms static portfolio data into a dynamic "physics-based" simulation, allowing users to identify optimal liquidation windows and stress-test the portfolio against macroeconomic shocks.

🚀 Core Features

- **Dynamic Portfolio Migration:** Watch assets drift from "Stellar" (high health) to "Red Zone" (delinquent) over a 24-month forecast horizon.
- **Stochastic Risk Modeling:** Every account has individualized "decay velocity," preventing unrealistic clustering and reflecting true portfolio heterogeneity.
- **"What-If" Scenario Engine:**
  - **Interest Rate Shock:** Real-time precision input to accelerate asset degradation based on rate hikes.
  - **Liquidation Boundary:** Adjustable "Sell Zone" threshold to define when an asset is no longer worth holding.
- **Financial Intelligence:** Live KPI cards calculate total Portfolio NPV, at-risk capital, and estimated cash recovery based on a variable Recovery Rate.
- **Advanced Visualizations:**
  - **Animated Bubble Chart:** Tracks individual account paths and value decay.
  - **Density Heatmap:** Identifies "toxic" risk concentrations in large-scale portfolios.

- **Self-Service Data:** Support for real-world data injection via CSV or Excel (XLSX) uploader.

🛠️ Installation & Setup

1. **Clone the Repository:**

```
git clone https://github.com
cd portfolio-strategy-dashboard
```

2. **Install Dependencies:** Ensure you have Python 3.8+ installed. Run:

```
pip install -r requirements.txt
```

3. **Run the Application:**

```
streamlit run app.py
```

📋 Requirements (requirements.txt)

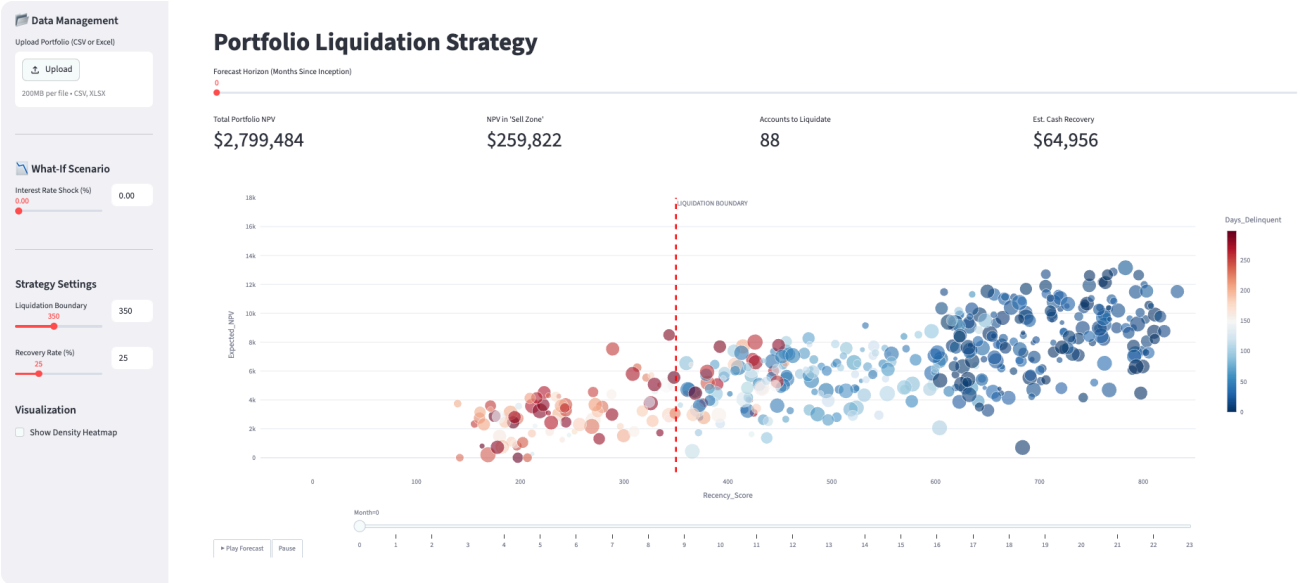
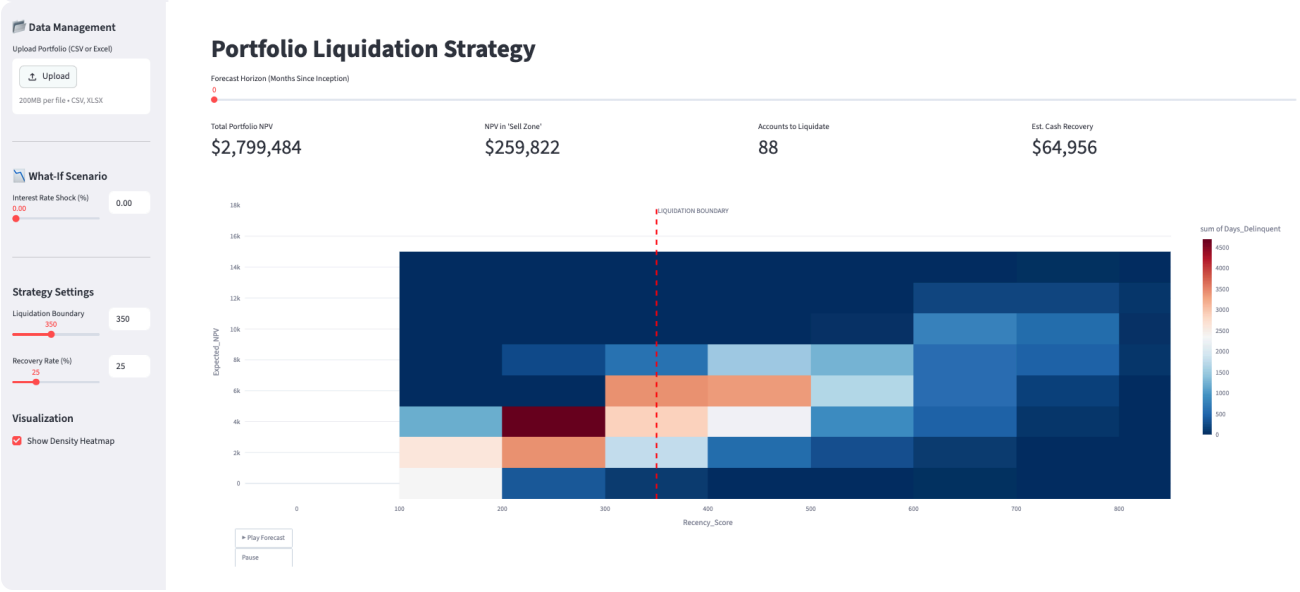
To ensure full functionality (including Excel support and high-performance visuals), the following packages are required:

```
streamlit
pandas
plotly
openpyxl
numpy
```

📈 Strategic Use Case

This tool is built to solve the "Cost of Waiting" dilemma. By simulating the aging process, asset managers can identify the "Sweet Spot" for liquidation—where the asset still retains meaningful market value (NPV) before the probability of default (Recency Score) collapses the recovery potential.

Disclaimer:

This tool is a prototype designed for simulation and strategic visualization purposes. Financial decisions should always be verified against internal risk models and professional auditing standards.

Screenshots: